

VS Code to Amazon S3

Setup, connection, upload and verification guide

Project: consumer-complaints-ml

Region: eu-west-2 | Bucket: stream-intel-lakehouse-107721655962-eu-west-2-an

Purpose. Use this guide whenever you need to connect a VS Code PowerShell terminal to AWS, verify access to Amazon S3, and upload data safely from Python or the AWS CLI.

1. What is installed and what each item does

Component	Purpose	Used in this project
VS Code	Write and manage project files	Python scripts, terminal and project folders
Python 3.11 virtual environment	Keeps project packages isolated	Located at .venv\Scripts\python.exe
boto3	Python SDK for AWS	Uploads and verifies files in S3
requests	Downloads files from public URLs	Downloads the CFPB ZIP file
AWS CLI	Runs AWS commands from PowerShell	Lists buckets, checks identity and verifies uploads
Amazon S3	Cloud object storage	Stores raw datasets and later model artefacts

2. One-time setup

2.1 Open the project folder

```
cd C:\Users\allif\consumer-complaints-ml
code .
```

- The VS Code Explorer should display .venv, data and ingestion folders.

2.2 Create the Python 3.11 environment

```
& "C:\Users\allif\AppData\Local\Programs\Python\Python311\python.exe" -m venv .venv
.venv\Scripts\Activate.ps1
```

```
python --version
python -c "import sys; print(sys.executable)"
```

- Expected interpreter: C:\Users\allif\consumer-complaints-ml\.venv\Scripts\python.exe

2.3 Install Python packages

```
python -m pip install --upgrade pip
python -m pip install boto3 requests pytest
python -c "import boto3, requests; print('Packages installed successfully')"
```

- Use python -m pip so packages are installed into the active .venv.

2.4 Make the AWS CLI visible to VS Code

```
$env:Path += ";C:\Program Files\Amazon\AWSCLIV2"
aws --version
```

- This does not install or authenticate AWS. It only lets the current PowerShell session locate aws.exe.

3. Configure AWS credentials

Security note: Never paste an AWS access key or secret access key into chat, source code, GitHub, screenshots or README files.

- In AWS Console: IAM > Users > fred-admin > Security credentials > Create access key.
- Choose Command Line Interface (CLI), save the Access Key ID and Secret Access Key securely, then run:

```
aws configure
```

```
AWS Access Key ID: <your access key ID>
AWS Secret Access Key: <your secret access key>
Default region name: eu-west-2
Default output format: json
```

- You normally run aws configure once. Credentials are stored in your local Windows user profile.

4. Standard connection sequence for every new session

Use this sequence each time you open a new VS Code PowerShell terminal.

```
cd C:\Users\allif\consumer-complaints-ml

.venv\Scripts\Activate.ps1

$env:Path += ";C:\Program Files\Amazon\AWSCLIV2"

aws sts get-caller-identity

$env:S3_BUCKET_NAME="stream-intel-lakehouse-107721655962-eu-west-2-an"

echo $env:S3_BUCKET_NAME

aws s3 ls "s3://$env:S3_BUCKET_NAME/"
```

- aws sts get-caller-identity confirms which AWS user/account is active.
- S3_BUCKET_NAME is a session variable and must be set again in a new terminal unless you make it permanent.

5. Test S3 access before a large upload

```
"test upload from VS Code" | Out-File s3_test.txt

aws s3 cp s3_test.txt "s3://$env:S3_BUCKET_NAME/test/s3_test.txt"

aws s3 ls "s3://$env:S3_BUCKET_NAME/test/"

aws s3 rm "s3://$env:S3_BUCKET_NAME/test/s3_test.txt"

Remove-Item s3_test.txt
```

- If upload, list and delete all succeed, your permissions are suitable for the project.

6. Upload data using Python

- Store the script at ingestion\consumer_complaints\download_to_s3.py.
- The script downloads the official CFPB complaints ZIP, uploads it with boto3 and verifies it with head_object.

```
echo $env:S3_BUCKET_NAME
python ingestion\consumer_complaints\download_to_s3.py
```

- Leave the terminal open until the script reports download complete, S3 upload complete and upload verification successful.

7. Verify the uploaded object

```
aws s3 ls "s3://$env:S3_BUCKET_NAME/raw/consumer_complaints/" --recursive --human-readable
```

- Expected object path:
raw/consumer_complaints/ingestion_date=YYYY-MM-DD/complaints.csv.zip
- The initial file uploaded on 30 July 2026 was approximately 1.3 GiB.

8. View the object in the AWS Console

1. Open AWS Console and search for S3.
2. Open stream-intel-lakehouse-107721655962-eu-west-2-an.
3. Open raw.
4. Open consumer_complaints.
5. Open ingestion_date=2026-07-30.
6. Confirm complaints.csv.zip is present.

9. Common errors and fixes

Error	Fix
'aws' is not recognised	\$env:Path += ";C:\Program Files\Amazon\AWSCLIV2" and then run aws --version.
Unable to locate credentials	Run aws configure, then aws sts get-caller-identity.
AccessDenied when uploading	Check the IAM user has permission for the target bucket and prefix.
S3_BUCKET_NAME is empty	Run \$env:S3_BUCKET_NAME="stream-intel-lakehouse-107721655962-eu-west-2-an".
No module named boto3/requests/pytest	Activate .venv, then run python -m pip install <package>.
Both (.venv) and (base) are displayed	The .venv is still used if sys.executable points to .venv. Optionally run conda deactivate.

10. Project safety and maintenance

- Keep data/, *.zip, .venv/, .env and __pycache__/ in .gitignore.
- Do not commit credentials or large datasets to GitHub.
- Use dated S3 prefixes so future snapshots do not overwrite earlier data.
- Deactivate and rotate any access key that may have been exposed.
- For a production environment, prefer short-lived credentials or IAM roles over long-lived access keys.
- After ingestion, the next stage is Databricks Bronze, Silver and Gold processing, followed by MLflow training and model export to S3.

Quick reference

```
# Open and activate
cd C:\Users\allif\consumer-complaints-ml
.venv\Scripts\Activate.ps1

# Make AWS CLI available
$env:Path += ";C:\Program Files\Amazon\AWSCLIV2"

# Confirm AWS access
aws sts get-caller-identity

# Select bucket
$env:S3_BUCKET_NAME="stream-intel-lakehouse-107721655962-eu-west-2-an"

# Check bucket
aws s3 ls "s3://$env:S3_BUCKET_NAME/"

# Run ingestion
python ingestion\consumer_complaints\download_to_s3.py

# Verify uploaded data
aws s3 ls "s3://$env:S3_BUCKET_NAME/raw/consumer_complaints/" --recursive --human-readable
```